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Client Entity: SWK Properties LLC (Dubai, UAE)
Live OS URL: https://os.swkpropertiesuae.com
Handover Portal: https://os.swkpropertiesuae.com/handover

DOC-02 OF 04 · TECHNICAL ARCHITECTURE MANUAL · CLOUD, DATABASE & SECURITY SPECIFICATION

SWK Properties LLC

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DOCUMENT 02 · TECHNICAL ARCHITECTURE, SECURITY & CLOUD INFRASTRUCTURE MANUAL

System Topology, Dual-Database Engine, RBAC & Cloudflare Runbook

Complete engineering reference for maintaining, operating, backing up, and extending **SWK OS** on Cloudflare Workers, Cloudflare Containers, and managed PostgreSQL 16.

1. 4-Tier Cloud-Native Architecture

Tier 1: Progressive Web App & Print Client

Built with **React 19 + TypeScript 5.9 + Vite 7** (`src/web/`). Includes an offline-resilient Service Worker (`public/sw.js`), Web Push notifications, bilingual English/Arabic RTL support, and a dedicated A4 portrait print engine.

Tier 2: Cloudflare Edge Worker & Container

Traffic to `os.swkpropertiesuae.com` terminates at Cloudflare Edge Worker `swk-os`, which orchestrates the `SwkOs` Durable Object Container (`node:22-bookworm-slim` with Chromium, Poppler & Tesseract OCR).

Tier 3: Fastify 5 Application & Security Core

High-throughput Node.js 22 server (`src/server/app.ts`) exposing **317 REST & webhook endpoints** with schema validation, optimistic locking, rate limiting, CSRF protection, and automated accounting/compliance hooks.

Tier 4: Dual PostgreSQL 16 + SQLite WAL Persistence

Implemented in `src/server/db.ts` and `src/server/pg-sync.ts`. Synchronizes all 75+ tables between local high-speed SQLite WAL storage and managed Cloud PostgreSQL (`DATABASE_URL`, schema `swk`).

2. Cryptographic Security, Audit Ledger & Document Standards

SECURITY / GOVERNANCE LAYER	IMPLEMENTATION FILE	ENGINEERING SPECIFICATION
Password Cryptography	`src/server/auth.ts`	Node.js `crypto.scrypt` with unique 16-byte random salts and `crypto.timingSafeEqual` verification.
Two-Factor Auth (2FA)	`src/server/auth.ts`	RFC 6238 TOTP (SHA-1, 6-digit, 30-second window) with single-use emergency recovery codes.

SECURITY / GOVERNANCE LAYER	IMPLEMENTATION FILE	ENGINEERING SPECIFICATION
Hard Staff Access Guard	`src/shared/swk-staff-access.ts`	Enforces `canPrintReceptionQr` (Owner & Haseeb only), `canAccessSwkAdminLogic`, and SQL self-scoping on staff HR records.
Tamper-Evident Audit Chain	`src/server/audit.ts`	Every mutation logs before/after JSON diffs chained via `SHA-256(prev_hash + payload)` to detect any tampering.
Corporate Document Standard	`src/shared/corporate-footer.ts` `src/web/print/PrintPage.tsx`	All printed documents strictly display main logo `/brand/swk-properties-horizontal.svg`, legal entity **SWK Properties LLC**, and **www.swk.properties · www.swkpropertiesuae.com · www.swk.rent**.

3. Production Deployment & CLI Maintenance Commands

```
# Verify TypeScript types, build production bundles & run automated tests:
npm ci && npm run typecheck && npm run build
npx tsx tests/company-documents.ts

# Automatic CI/CD Deployment to https://os.swkpropertiesuae.com:
git push origin main   # Triggers .github/workflows/deploy-cloudflare.yml
```